WDG3A - 3100 HP GOODS LOCOMOTIVE

Main line, diesel electric goods traffic locomotive equipped with 16 cylinder Alco engine and microprocessor based traction control system.

The locomotive is powered by BLW make 251-C up-rated 16 cylinder fuel efficient turbo-supercharged engine with a large after cooler for better cooling of engine inlet air. The engine develops 3100 hp at 1050 rpm under standard conditions.

The locomotive is provided with AC-DC microprocessor based traction control system consisting of a directly coupled self ventilated BHEL make TA10102 EV traction alternator.

Apart from Indian Railways, this loco is used by a large number of Indian Non - Railways Customers such as power Plants, Port Trusts etc.

Diesel Engine

- 16 Cylinder 251 C Engine
  4 Stroke, turbocharged
- Fuel Efficient Engine
- Injection System - Direct Injection
- Governor - MCBG
- Compression Ratio - 11.75:1
- Lube Oil Sump Capacity - 1025 Ltrs

Brakes

- Panel Mounted IRAB-1 Brake system
- Air, hand, Dynamic brake
- Pure air brake

Transmission

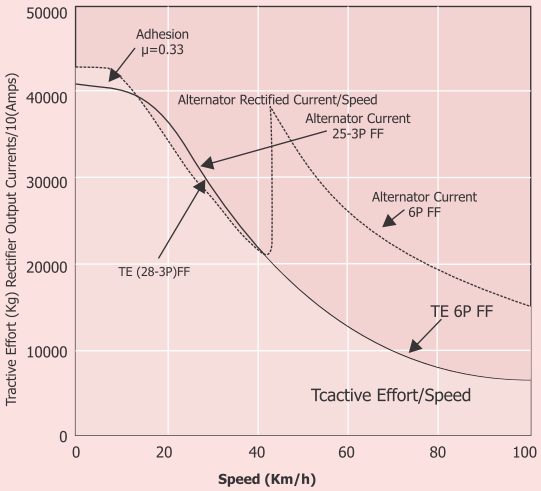
- Electrical AC-DC
- Microprocessor Control
  6 Traction motor
  (3 per bogie)
- Suspension - Axle hung / nose suspension
- Gear Ratio - 18 : 74

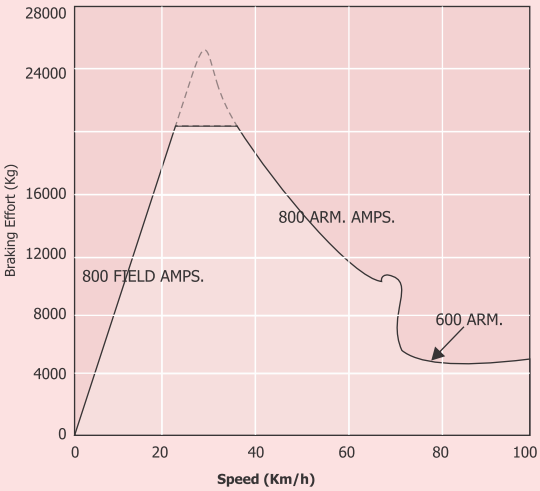
Truck

- High adhesion Fabricated Co-Co Bogie
  Adhesion (μ) - 0.33
- Unidirectional Traction motor arrangement for high adhesion

General Characteristics	
Installed Power	3100 HP
Axle Load	20.5 T
Gauge	1676 mm
Wheel arrangement	Co-Co
Wheel diameter	1092 mm
Height	4162 mm
Width	3016 mm
Overall Length (Over Buffer Beam)	17850 mm
Weight	123 T
Max tractive effort	40.5 T
Maximum Speed	100 Kmph
Fuel tank capacity	6000 lts
Locomotive Control	Micro-processor Control

Tractive Effort Chart





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